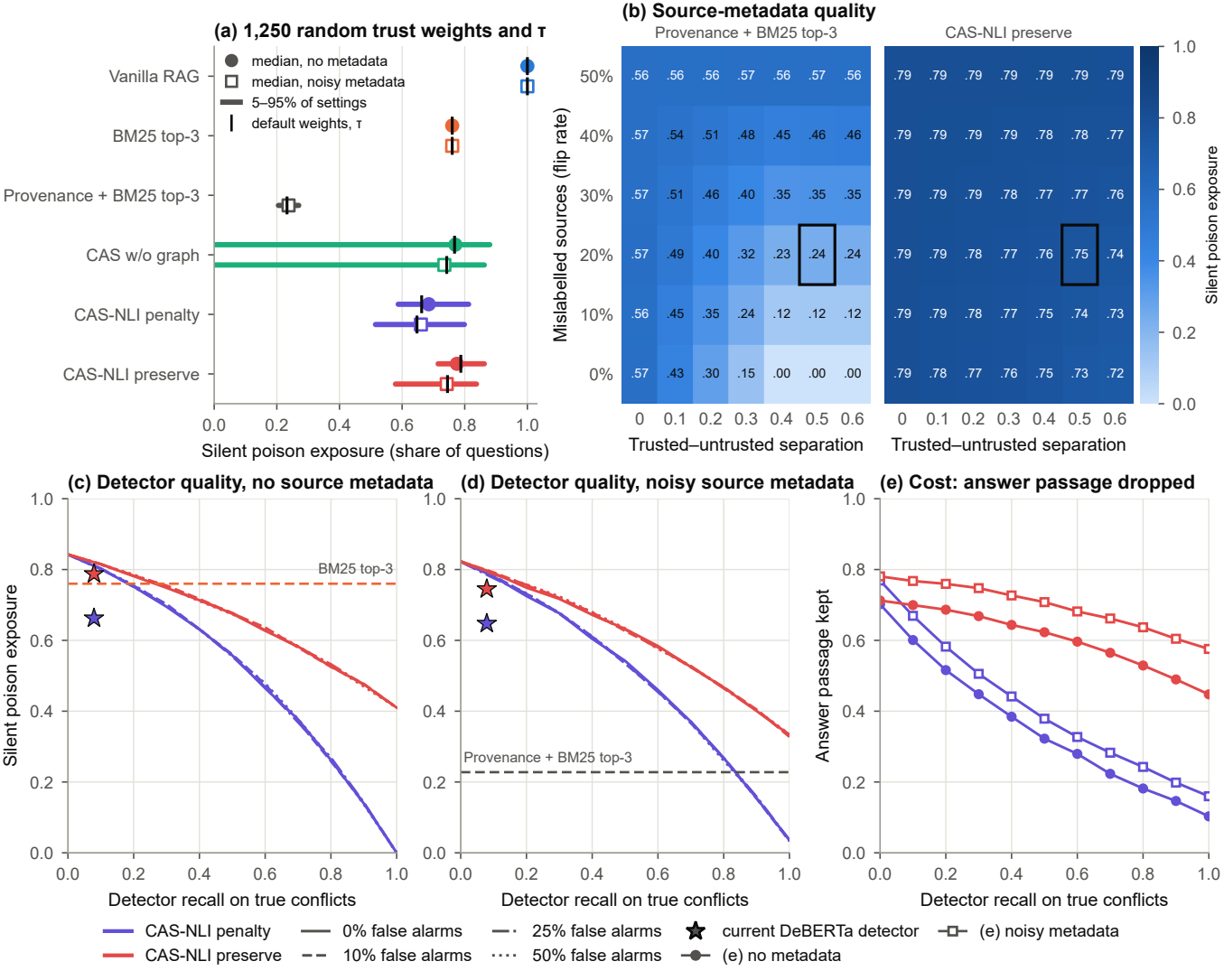

(a) 1,250 random trust weights and τ
median, no metadata
median, noisy metadata
5–95% of settings
default weights, τ
Vanilla RAG
BM25 top-3
Provenance + BM25 top-3
CAS w/o graph
CAS-NLI penalty
CAS-NLI preserve
0.0 0.2 0.4 0.6 0.8 1.0
Silent poison exposure (share of questions)
(b) Source-metadata quality
Provenance + BM25 top-3
CAS-NLI preserve
Misabelled sources (flip rate)
50% .56 .56 .56 .57 .56 .57 .56 .79 .79 .79 .79 .79 .79 .79
40% .57 .54 .51 .48 .45 .46 .46 .79 .79 .79 .79 .78 .78 .77
30% .57 .51 .46 .40 .35 .35 .35 .79 .79 .79 .78 .77 .77 .76
20% .57 .49 .40 .32 .23 .24 .24 .79 .79 .78 .77 .76 .75 .74
10% .56 .45 .35 .24 .12 .12 .12 .79 .79 .78 .77 .75 .74 .73
0% .57 .43 .30 .15 .00 .00 .00 .79 .78 .77 .76 .75 .73 .72
0 0.1 0.2 0.3 0.4 0.5 0.6
Trusted–untrusted separation
Silent poison exposure
1.0 0.8 0.6 0.4 0.2 0.0
(c) Detector quality, no source metadata
(d) Detector quality, noisy source metadata
(e) Cost: answer passage dropped
Silent poison exposure
Answer passage kept
BM25 top-3
Provenance + BM25 top-3
Detector recall on true conflicts
CAS-NLI penalty
CAS-NLI preserve
0% false alarms
10% false alarms
25% false alarms
50% false alarms
current DeBERTa detector
(e) no metadata
(e) noisy metadata